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Source: *Sankhyā: The Indian Journal of Statistics, Series B (1960-2002)*, Vol. 41, No. 3/4 (Dec., 1979), pp. 154-169

Published by: [Springer](#) on behalf of the [Indian Statistical Institute](#)

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Accessed: 15/10/2013 12:48

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DISTRIBUTION OF THE LIKELIHOOD RATIO STATISTIC FOR TESTING SPHERICITY STRUCTURE FOR A NORMAL COVARIANCE MATRIX AND ITS PERCENTAGE POINTS

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SUMMARY. The null distribution of the likelihood ratio criterion for testing sphericity structure in a normal p -variate covariance matrix Σ (i.e. for testing the hypothesis that all off diagonal elements of Σ are zero while the diagonal elements are equal, in sets) has been obtained in series form and percentage points computed for small p , $\alpha = 0.05, 0.01$, and various values of sample size, large as well as small. Comparison is also made with the approximate values using Box's series.

1. INTRODUCTION

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be independent normal random p -vectors with mean vector μ and positive definite covariance matrix Σ , i.e.,

$$\mathbf{x}_i \sim N(\mu, \Sigma), \quad \Sigma > 0.$$

Let

$$\bar{\mathbf{x}} = N^{-1} \sum_{i=1}^N \mathbf{x}_i$$

and

$$\mathbf{S} = \sum_{i=1}^N (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})', \quad n = N-1. \quad \dots \quad (1.1)$$

To test the hypothesis of sphericity, namely, $H_0 : \Sigma = \sigma^2 \mathbf{I}_p$, where $\sigma^2 > 0$ is unknown, against $H_1 : \Sigma \neq \sigma^2 \mathbf{I}_p$, Mauchly (1940b) obtained the likelihood ratio test criterion for H_0 in the form $W = |\mathbf{S}| / [(\text{tr } \mathbf{S})/p]^p$. In this paper we wish to consider a more general problem of testing a sphericity structure for the covariance matrix Σ i.e., we wish to test the hypothesis that all off diagonal elements of Σ are zero and the diagonal elements are equal in sets.

Mauchly's sphericity test stated above and Wilks' L_{VC} criterion are special cases of such a hypothesis. In Section 2, we derive the likelihood ratio criterion for this problem and obtain its moments. Because of the importance of this criterion (see Abruzzi, 1950; Mauchly, 1940a; Cole and Grizzle, 1966; Hickman, 1953), suitable methods are developed to obtain the exact distribution of the likelihood ratio criterion in series forms, from which exact percentage points of the criterion are obtainable for any desired N . Tables of 5% and 1% significance levels are given for various p and N and comparisons made with approximate values using Box's series (Anderson, 1958, p. 263).

2. THE LIKELIHOOD RATIO CRITERION AND ITS MOMENTS

In this section we obtain the likelihood ratio test of the hypothesis

$$H_0 : \Sigma = \begin{bmatrix} \sigma_1^2 \mathbf{I}_{p_1} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \sigma_2^2 \mathbf{I}_{p_2} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \sigma_q^2 \mathbf{I}_{p_q} \end{bmatrix}, \sigma_i^2 > 0 \text{ unknown, } \dots \quad (2.1)$$

against the general alternatives, where $\mathbf{I}_{p_i}$ is the identity matrix of order p_i and $p_1 + p_2 + \dots + p_q = p$.

Let

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} & \dots & \mathbf{S}_{1q} \\ \mathbf{S}_{21} & \mathbf{S}_{22} & \dots & \mathbf{S}_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{S}_{q1} & \dots & \dots & \mathbf{S}_{qq} \end{bmatrix}$$

be partitioned similar to (2.1). Then it can be easily seen that the likelihood ratio criterion for testing H_0 against general alternatives $H \neq H_0$ is based on the statistic

$$\lambda = |\mathbf{S}|^{N/2} \left/ \left\{ \prod_{i=1}^q [(\text{tr } \mathbf{S}_{ii}/p_i)^{p_i}]^{N/2} \right\} \right. \dots \quad (2.2)$$

Using equation (16), page 262 of Anderson (1958) for the moments of the Mauchly's sphericity criterion, the moments of $W = \lambda^{2/N}$ is given by

$$E(W^h) = \prod_{i=1}^q \left\{ p_i^{hp_i} \frac{\Gamma\left(\frac{np_i}{2}\right)}{\Gamma\left(\frac{np_i}{2} + hp_i\right)} \right\} \prod_{\alpha=1}^p \frac{\Gamma\left\{\frac{1}{2}(n+1-\alpha)+h\right\}}{\Gamma\left\{\frac{1}{2}(n+1-\alpha)\right\}} \quad \dots \quad (2.3)$$

where $n = N-1$.

Special cases : (1) If $p_1 = p$ and $q = 1$, then the moments of W reduce to those of Mauchly's sphericity test and the hypothesis H_0 in (2.1) is the usual sphericity test.

(2) If $p_1 = 1$, $p_2 = p-1$ and $q = 2$, then the moments in (2.3) agree with the moments of Wilks' L_{VC} criterion (see Wilks, 1946) and thus the distribution of W is that of L_{VC} in this case.

3. EXACT DISTRIBUTION OF W

In part (a) of this section we obtain the distribution of W as a beta series while in part (b) we obtain its distribution as a chi-square series. It is to be noted that the relative merits of methods (a), (b) lie in the fact that when N is small the former is more efficient than the latter in the sense that it only needs considerably less number of terms of the series to attain the desired degree of accuracy for computing the percentage points while the latter is more efficient than the former when N is moderately large.

(a) *Distribution of W as a Beta series.* From (2.3) on taking inverse Mellin transform and following the method as described in Nagarsenker and Das (1975), Nagarsenker and Pillai (1972), the density of W is given by

$$f(W) = K_1(p_i, N, W) \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} W^{-s} s^{-v} \left[1 + \sum_{r=1}^{\infty} (A_r/s^r) \right] ds, \quad \dots \quad (3.1)$$

where the coefficients A_r are available in Nagarsenker and Pillai (1972) and also the first few coefficients which are needed in the computations are given in the appendix for completeness,

$$K_1(p_i, N, W) = (2\pi)^{(p-q)/2} W^{\frac{1}{2}(N-\mu)-1} \left\{ \prod_{i=1}^q \Gamma\left(\frac{np_i}{2}\right) \right\} \left/ \prod_{i=1}^p \Gamma\left[\frac{1}{2}(N-j)\right] \right\} \\ \cdot \left(\prod_{i=1}^q p_i^{-\frac{1}{2}(N-\mu)p_i} \right) \left/ \prod_{i=1}^q (p_i)^{((\mu-1)p_i-1)/2} \right.$$

where $v = (p^2 + p - 2q)/4$, $c = \frac{1}{2}(N - \mu)$ and μ is an adjustable constant which can be chosen to govern the rate of convergence of the resulting Beta series. Here μ may be conveniently arbitrarily chosen and adjusted through the use of computer so that the value of the integral over zero to one of the resulting series is close to one. Alternately it can be chosen so that the coefficient $A_1 = 0$ and this leads to the choice of μ given by

$$\mu = \frac{2p^3 + 9p^2 + 5p - 12q + 4 \sum_{i=1}^q (1/p_i)}{6(p^2 + p - 2q)}.$$

This value of μ also gives the best means of initial guess of its value to be used in the first method of the choice of μ .

The integral on the right hand side of (3.1) can be easily computed if v is an integer and its value is the residue of

$$W^{-s} s^{-v} \left[1 + \sum_{r=1}^{\infty} (A_r/s^r) \right] \text{ at } s = 0$$

which is given by

$$\sum_{r=0}^{\infty} [A_r (-\log W)^{v+r-1} / \Gamma(v+r)], \quad A_0 = 1 \quad \dots \quad (3.2)$$

and thus from (3.1), the density of W is

$$f(W) = K_1(p_i, N, W) \sum_{r=0}^{\infty} [A_r (-\log W)^{v+r-1} / \Gamma(v+r)]. \quad \dots \quad (3.3)$$

Proceeding as in Nagarsenker and Das (1975) and writing $K_2(p_i, N) = K_1(p_i, N, W)/W^{\frac{1}{2}(N-\mu)-1}$ the c.d.f of W is obtained as

$$P(W \leq x_0) = K_2(p_i, N) \sum_{r=0}^{\infty} (A_r) I_{v+r-1, u}(x_0), \quad \dots \quad (3.4)$$

where

$$I_{v+r-1,\mu}(x_0) = \int_0^{x_0} W^u (-\log W)^{v+r-1} dW / \Gamma(v+r)$$

and satisfies the recurrence relation (2.16) in Nagarsenker and Pillai (1973). It is to be noted that (3.4) holds only if v is an integer. However, if v is not an integer using a theorem due to Nair (1940), we can expand as a factorial series the integrand in (3.1) since,

$$s^{-v} \left[1 + \sum_{r=1}^{\infty} (A_r/s^r) \right] = \sum_{i=0}^{\infty} R_i \Gamma(s) / \Gamma(s+v+i), \quad \dots \quad (3.5)$$

R_i 's being available in Nagarsenker and Pillai (1972) and some of these which are needed in the computations are given in the appendix. Using (3.5) in (3.1) and integrating term by term, we have for integral v the c.d.f of W in the form

$$p(W \leq x_0) = K_2(p_i, N) \sum_{i=0}^{\infty} (R_i) I_{x_0} \left(\frac{1}{2} (N-\mu), v+i \right) / \Gamma(v+i), \quad \dots \quad (3.6)$$

where $I_{x_0}(p, q)$ is the incomplete beta function $\int_0^{x_0} x^{p-1} (1-x)^{q-1} dx$. It is to be noted that for $p = 2$, we have proceeding as in Nagarsenker and Das (1975), the following alternate expression for the density of W

$$f(W) = \Gamma\left(\frac{n}{2}\right) W^{\frac{1}{2}(n-1)-1} (1-W)^{-\frac{1}{2}} / \sqrt{\pi} \Gamma\left(\frac{n-1}{2}\right). \quad \dots \quad (3.7)$$

(b) *Distribution of W as a chisquare series.* We shall now obtain the distribution of W in chisquare series form. For this let $L = -(N-U)\log W$, where U is an adjustable constant similar to μ in (3.1). If $\phi(t)$ is the characteristic function of L then

$$\phi(t) = K(p_j, n) \cdot c(t), \quad \dots \quad (3.8)$$

where

$$c(t) = \left[\prod_{j=1}^q p_j^{-(N-U)p_j i t} / \Gamma \left\{ \left(\frac{N-1}{2} \right) p_j - (N-U)p_j i t \right\} \right] \prod_{\alpha=1}^p \Gamma \left\{ \frac{1}{2} (N-\alpha) - (N-U) i t \right\} \quad \dots \quad (3.9)$$

and

$$K(p_j, n) = \prod_{j=1}^q \Gamma(np_j/2) / \prod_{\alpha=1}^p \Gamma\left\{\frac{1}{2}(n+1-\alpha)\right\}.$$

Using an expansion for $\log \Gamma(x+h)$ given in (2.7) of Nagarsenker and Pillai (1973), the characteristic function L can be obtained from (3.8) as

$$\phi(t) = K_1(p_j, n)[(N-U)(1-2it)/2]^{-v} \left[\sum_{j=0}^{\infty} B_j / ((N-U)(1-2it)/2)^j \right] + R_{m+1}(n, t), \quad \dots \quad (3.10)$$

where

$$K_1(p_j, n) = K(p_j, n)(2\pi)^{(p-q)/2} \left(\prod_{j=1}^q p_j^{-\frac{(N-1)}{2} p_j + 1/2} \right)$$

and the coefficients B_j 's are given in the appendix. Hence the density of L is given by

$$f(L) = K_1(p_j, n) \sum_{j=0}^{\infty} (B_j) \left(\frac{2}{N-U} \right)^{j+v} g_{2j+2v}(\chi^2) + R_{m+1}''(n), \quad \dots \quad (3.11)$$

where $g_{\alpha}(\chi^2)$ is the chisquare density with α d.f. Hence the probability that L is larger than any value, say x_0 , is

$$P(L \geq x_0) = K_1(p_j, n) \sum_{j=0}^{\infty} (B_j) \left(\frac{2}{N-U} \right)^{j+v} G_{2j+2v}(\chi^2) + R_{m+1}(N), \quad \dots \quad (3.12)$$

where

$$G_{2j+2v}(\chi^2) = \int_{x_0}^{\infty} g_{2j+2v}(\chi^2) d\chi^2$$

and

$$R_{m+1}(N) = \frac{1}{2\pi} K_1(p_i, n) \int_{x_0}^{\infty} \int_{-\infty}^{\infty} e^{-itL} \sum_{j=0}^{\infty} (B_j) \left(\frac{2}{N-U} \right)^{j+v} \cdot (1-2it)^{-(j+v)} [\exp\{R_{m+1}(N, t)\} - 1] dt dL.$$

Thus from (3.12) we have that the distribution of L may be obtained as a series of chisquare distributions.

4. APPROXIMATION USING BOX'S SERIES

Let $\lambda = W^{n/2}$. Then from (2.3), the h -th moment of λ is

$$E(\lambda^h) = K(p_i, n) \left[\prod_{i=1}^q p_i^{np_i h/2} / \Gamma\{np_i(1+h)/2\} \right] \\ \cdot \prod_{\alpha=1}^p \Gamma \left\{ \frac{n}{2} (1+h) + \frac{1}{2} (1-\alpha) \right\} \quad \dots \quad (4.1)$$

and this is of the form given in equation (1), page 203 of Anderson (1958) with $x_k = n/2$, $y_j = np_j/2$, $\xi_k = (1-k)/2$, $n_j = 0$, $a = p$, $b = q$, $k = 1, 2, \dots, p$ and $j = 1, 2, \dots, q$. Thus following Anderson (1958), the asymptotic expansion for the c.d.f of $-n\rho \log W$ is given by

$$P(-n\rho \log W \leq z) = P(\chi_f^2 \leq z) + w_2(P\{\chi_{f+4}^2 \leq z\} - P\{\chi_f^2 \leq z\}) + O(n^{-3}), \quad \dots \quad (4.2)$$

where

$$f = \frac{p(p+1)}{2} - q, (1-\rho) = \frac{2p^3 + 3p^2 - p - 4 \sum_{i=1}^q (1/p_i)}{6n(p^2 + p - 2q)}$$

and

$$w_2 = \frac{1}{288(n\rho)^2} \left[2p^3 + 3p^2 - p - 4 \left(\sum_{i=1}^q 1/p_i \right) \right]^2 / (p^2 + p - 2q) \\ + \left\{ \left[2p^3 + 3p^2 - p - 4 \left(\sum_{i=1}^q 1/p_i \right) \right] / (p^2 + p - 2q) \right\} \\ \cdot \left\{ -4p^3 - 6p^2 + 2p + 8 \left(\sum_{j=1}^q 1/p_j \right) \right\} + (6p^4 + 12p^3 - 6p^2 - 12p).$$

For the special case in which $p = q$ and each $p_j = 1$, we have from (4.2) the asymptotic expansion for the c.d.f of $-n\rho \log W$ as

$$P(-n\rho \log W \leq z) = P(\chi_f^2 \leq z) + w_2(P\{\chi_{f+4}^2 \leq z\} - P\{\chi_f^2 \leq z\}) + O(n^{-3}), \quad \dots \quad (4.3)$$

where

$$f = p(p-1)/2, \quad \rho = 1 - (2p+5)/6n,$$

and

$$w_2 = p(p-1)(2p^2 - 2p - 13)/288n^2\rho^2.$$

5. COMPUTATIONS USING SERIES FORMS AND BOX'S APPROXIMATION

The 0.05 and 0.01 significance points of W were computed for various N in the following cases: (1) $p = q = 3(1)8$ and each $p_j = 1$ (2) $p = 3(1)5$ with different possible values of p_j and q . It is to be noted that the distribution of W for example with $p = 4, q = 3, p_1 = 2, p_2 = 1, p_3 = 1$ is the same as W with $p = 4, q = 3, p_1 = 1, p_2 = 2, p_3 = 1$ etc., that is any interchange of the values of p_j does not change the distribution of W for a given p and q . The computation was carried out on UNIVAC 1108 using double precision arithmetic and the values are given correct to five significant figures in Tables 2 to 5. The accuracy of the tabled values of the percentage points calculated from the series (3.4), (3.6) and (3.12) is tested using $p = 2$ in these series against its beta distribution in (3.7) and the results were seen to agree completely to the last significant figure. Table 1 compares the exact 0.05 significance points of W for the special case in which all $p_i = 1$ with those computed using Box's asymptotic expansion for $-n\rho \log W$ given in (4.3). Table 1 reveals that Box's series approximation is reasonably good for small values of p and moderate values of N . It is to be noted that Tables 2 and 3 can be used to test the hypothesis $H_0: \mathbf{P} = \mathbf{I}$ against the alternative $H_1: \mathbf{P} \neq \mathbf{I}$ where $\mathbf{P}$ denotes the population correlation matrix of normal multivariate distribution, since when each $p_j = 1$ and $p = q$, the criterion λ in (2.2) reduces to the criterion for testing this hypothesis.

TABLE 1. COMPARISON OF 5% POINTS OF W WHEN $p_i = 1$

N	$p = 3$		$p = 5$		$p = 8$	
	Box's	Exact	Box's	Exact	Box's	Exact
10	.33633	.33621	.056237	.055468	.0328816	.0312528
15	.52616	.52614	.20124	.20101	.017493	.016936
20	.63434	.63433	.32845	.32837	.067065	.056703
40	.81038	.81038	.60537	.60536	.31117	.31112
50	.84732	.84732	.67444	.67443	.40255	.40253
80	.90369	.90369	.78714	.78714	.57821	.57820
100	.92272	.92272	.82718	.82718	.64856	.64856

TABLE 2. 5% POINTS OF W WHEN $p_t = 1$

N^p	3	4	5	6	7	8
4	·0210183					
5	·026873	·0324025				
6	·084781	·0286215	·0466687			
7	·15341	·033427	·0229095	·0420033		
8	·22052	·070026	·013279	·0210067	·0561052	
9	·28177	·11213	·031479	·0252641	·0235344	·0519002
10	·33621	·15558	·055468	·013903	·0220781	·0312528
11	·38420	·19807	·083017	·026714	·0260434	·0381667
12	·42644	·23847	·11237	·042905	·012570	·0225909
13	·46372	·27630	·14231	·061570	·021567	·0257984
14	·49674	·31143	·17202	·081894	·032721	·010585
15	·52614	·34393	·20101	·10321	·045639	·016936
16	·55243	·37393	·22896	·12502	·059925	·024731
17	·57605	·40162	·25572	·14692	·075219	·033798
18	·59738	·42720	·28121	·16866	·091212	·043943
19	·61672	·45087	·30542	·19002	·10765	·054971
20	·63433	·47278	·32837	·21089	·12431	·066703
25	·70291	·56146	·42600	·30562	·20651	·13089
30	·75003	·62530	·50067	·38381	·28099	·19598
35	·78432	·67320	·55892	·44781	·34564	·25658
40	·81038	·71038	·60536	·50057	·40112	·31112
45	·83083	·74003	·64315	·54456	·44876	·35959
50	·84732	·76421	·67444	·58167	·48986	·40253
60	·87223	·80125	·72317	·64066	·55677	·47444
70	·89017	·82827	·75930	·68529	·60863	·53174
80	·90369	·84883	·78714	·72017	·64986	·57820
90	·91425	·86501	·80923	·74815	·68335	·61650
100	·92272	·87806	·82718	·77108	·71106	·64856

TABLE 3. 1% POINTS OF W WHEN $p_t = 1$

N^p	3	4	5	6	7	8
4	.0437877					
5	.0251559	.0583414				
6	.027706	.0215806	.0524479			
7	.065713	.010309	.0351475	.0679738		
8	.11136	.028229	.0239086	.0317298	.0623717	
9	.15898	.053352	.012072	.0214901	.0459233	.0769806
10	.20548	.082925	.025106	.0251096	.0356880	.0420544
11	.24940	.11470	.042152	.011580	.0221402	.0321704
12	.29018	.14713	.062115	.020879	.0252458	.0388792
13	.32772	.17922	.083991	.032659	.010106	.0323389
14	.36214	.21038	.10697	.046453	.016726	.0247941
15	.39366	.24026	.13044	.061784	.024976	.0283719
16	.42254	.26870	.15895	.078223	.034651	.013099
17	.44902	.29564	.17719	.095402	.045521	.018933
18	.47336	.32109	.19994	.11302	.057356	.025788
19	.49576	.34508	.22207	.13085	.069945	.033555
20	.51644	.36769	.24348	.14870	.083097	.042115
25	.59944	.46263	.33884	.23436	.15243	.092832
30	.65864	.53421	.41592	.30979	.22016	.14891
35	.70280	.58956	.47824	.37417	.28185	.20406
40	.73695	.63344	.52919	.42883	.33661	.25563
45	.76411	.66899	.57142	.47539	.38480	.30277
50	.78622	.69834	.60690	.51534	.42718	.34543
60	.82000	.74389	.66303	.57999	.49763	.41857
70	.84459	.57	.70532	.62981	.55336	.47821
80	.86328	.80346	.73827	.66925	.59832	.52737
90	.87796	.82397	.76464	.70120	.63524	.56839
100	.88980	.84061	.78620	.72758	.66606	.60306

TABLE 4. 5% POINTS OF W WHEN p_i 's ARE NOT ALL EQUAL

N	$p=3, q=2$	$p=4$	$q=2$	$p=4, q=3$	$p=5,$	$q=2$
	$p_1=2, p_2=1$	$p_1=3, p_2=1$	$p_1=2, p_2=2$	$p_1=2$ $p_2=p_3=1$	$p_1=4, p_2=1$	$p_1=3, p_2=2$
4	·0364127					
5	·018320	·0312635	·0312954	·0317593		
6	·061804	·0250303	·0249993	·0265619	·0432727	·0433004
7	·11737	·021109	·021011	·026482	·0215517	·0215439
8	·17489	·046971	·046802	·057198	·0276651	·0276349
9	·22973	·078826	·078602	·093794	·019355	·019294
10	·28018	·11353	·11327	·13262	·035879	·035788
11	·32589	·14898	·14870	·17146	·055972	·055855
12	·36704	·18392	·18363	·20907	·078419	·078281
13	·40404	·21763	·21734	·24483	·10223	·10208
14	·43734	·24974	·24945	·27848	·12667	·12651
15	·46739	·28009	·27980	·30995	·15120	·15103
16	·49458	·30864	·30837	·33929	·17545	·17528
17	·51927	·33543	·33517	·36661	·19917	·19900
18	·54176	·36054	·36029	·39202	·22221	·22204
19	·56231	·38406	·38382	·41569	·24447	·24430
20	·58116	·40610	·40587	·43775	·26589	·26572
25	·65577	·49765	·49747	·52818	·36028	·36013
30	·70811	·56580	·56566	·59443	·43574	·43562
35	·74675	·61812	·61800	·64472	·49640	·49630
40	·77642	·65939	·65930	·68409	·54584	·54576
45	·79989	·69273	·69266	·71569	·58674	·58667
50	·81892	·72019	·72013	·74160	·62105	·62099
60	·84788	·76271	·76266	·78150	·67527	·67522
70	·86886	·79406	·79403	·81077	·71607	·71603
80	·88477	·81813	·81810	·83315	·74784	·74781
90	·89724	·83717	·83715	·85080	·77325	·77323
100	·90727	·85261	·85259	·86509	·79404	·79402

TABLE 4. (*Contd.*) 5% POINTS OF W WHEN p_i 's ARE NOT ALL EQUAL

	$p = 5,$ $p_1 = 3$ $p_2 = p_3 = 1$	$q = 3$ $p_1 = p_2 = 2$ $p_3 = 1$	$p = 5, q = 4$ $p_1 = 2$ $p_2 = p_3 = p_4 = 1$
4			
5			
6	·0441511	·0441427	·0452367
7	·0219090	·0219013	·0223515
8	·0291869	·0291580	·011025
9	·022720	·022663	·026701
10	·041417	·041333	·047865
11	·063736	·063628	·072653
12	·088288	·088163	·099499
13	·11400	·11387	·12725
14	·14011	·13996	·15511
15	·16607	·16592	·18256
16	·19153	·19138	·20925
17	·21626	·21611	·23500
18	·24014	·23999	·25970
19	·26307	·26292	·28328
20	·28503	·28489	·30576
25	·38072	·38060	·40255
30	·45614	·45604	·47772
35	·51618	·51610	·53697
40	·56477	·56470	·58457
45	·60475	·60469	·62352
50	·63816	·63811	·65592
60	·69069	·69065	·70663
70	·73003	·73000	·74442
80	·76056	·76054	·77365
90	·78493	·78491	·79691
100	·80481	·80479	·81584

TABLE 5. 1% POINTS OF W WHEN p_i 's ARE NOT ALL EQUAL

N	$p=3, q=2$	$p=4$	$q=2$	$p=4, q=3$	$p=5$	$q=2$
	$p_1=2, p_2=1$	$p_1=3, p_2=1$	$p_1=2, p_2=2$	$p_1=2$ $p_3=f_3=1$	$p_1=4, p_2=1$	$p_1=3, p_2=2$
4	·0425153					
5	·0234695	·0 ⁸ 43589	·0550608	·0564396		
6	·019843	·0390698	·0390119	·0211931	·0512499	·0513298
7	·049316	·0263672	·0263344	·0280765	·0327002	·0326860
8	·086620	·018484	·018408	·022779	·0222060	·0221965
9	·12719	·036601	·036481	·044079	·0272407	·0272154
10	·16815	·059078	·058919	·069833	·015834	·015788
11	·20790	·084300	·084110	·098127	·027717	·027650
12	·24567	·11098	·11077	·12754	·042300	·042215
13	·28109	·13820	·13797	·15710	·058928	·058826
14	·31411	·16531	·16508	·18618	·076996	·076880
15	·34477	·19191	·19167	·21439	·095996	·095870
16	·37320	·21772	·21748	·24151	·11552	·11539
17	·39956	·24258	·24235	·26743	·13526	·13512
18	·42401	·26643	·26620	·29210	·15497	·15483
19	·44671	·28921	·28899	·31552	·17448	·17434
20	·46782	·31094	·31072	·33772	·19367	·19352
25	·55403	·40477	·40459	·43232	·28233	·28219
30	·61690	·47807	·47792	·50495	·35739	·35727
35	·66453	·53614	·53602	·56183	·42005	·41994
40	·70177	·58300	·58290	·60733	·47248	·47239
45	·73164	·62148	·62139	·64445	·51672	·51664
50	·75612	·65359	·65351	·67526	·55441	·55434
60	·79380	·70401	·70396	·72339	·61496	·61490
70	·82143	·74174	·74170	·75918	·66130	·66126
80	·84255	·77100	·77096	·78682	·69782	·69779
90	·85921	·79433	·79430	·80879	·72731	·72728
100	·87269	·81336	·81334	·82667	·75159	·75157

TABLE 5 (Contd.) 1% POINTS OF W WHEN p_i 's ARE NOT ALL EQUAL

N	$p = 5,$ $p_1 = 3$ $p_2 = p_3 = 1$	$q = 3$ $p_1 = p_2 = 2$ $p_3 = 1$	$p = 5, q = 4$ $p_1 = 2$ $p_2 = p_3 = p_4 = 1$
4			
5			
6	.0515890	.0515997	.0519560
7	.0333401	.0333262	.0341369
8	.0226638	.0226547	.0232205
9	.0285692	.0285454	.010154
10	.018431	.018388	.012148
11	.031821	.031760	.036576
12	.048006	.047928	.054541
13	.066222	.066129	.074497
14	.085797	.085694	.095703
15	.10619	.10608	.11758
16	.12697	.12685	.13969
17	.14783	.14771	.16171
18	.16853	.16840	.18342
19	.18889	.18877	.20466
20	.20881	.20869	.22533
25	.29979	.29967	.31855
30	.37567	.37557	.39511
35	.43835	.43826	.45769
40	.49042	.49034	.50927
45	.53410	.53403	.55229
50	.57114	.57108	.58860
60	.63036	.63032	.64636
70	.67545	.67542	.69010
80	.71086	.71083	.72432
90	.73937	.73935	.75179
100	.76279	.76278	.77432

Appendix

(A) Coefficients A_r in (3.1) which have been used in the computations are :

$$\begin{aligned}
 A_0 &= 1, \\
 A_1 &= q_1, \\
 A_2 &= q_1^2/2 + q_2, \\
 A_3 &= q_3 + q_1q_2 + q_1^3/6, \\
 A_4 &= q_4 + q_3q_1 + q_2^2/2 + q_2q_1^2/2 + q_1^4/24, \\
 A_5 &= q_5 + q_4q_1 + q_3q_2 + q_3q_1^2/2 + q_2^2q_1/2 + q_1^3q_2/6 + q_1^5/120, \\
 A_6 &= q_6 + q_5q_1 + q_4q_2 + q_3^2/2 + q_4q_1^2/2 + q_3q_2q_1 + q_2^3/6 \\
 &\quad + q_1^3q_3/6 + q_1^2q_2^2/4 + q_1^4q_2/24 + q_1^6/720, \\
 A_7 &= q_7 + q_6q_1 + q_5q_2 + q_4q_3 + q_5q_1^2/2 + q_4q_2q_1 \\
 &\quad + q_3^2q_1/2 + q_3q_2^2/2 + q_1^3q_4/6 + q_1^2q_2q_3/2 + q_1q_2^3/6 \\
 &\quad + q_1^4q_3/24 + q_1^3q_2^2/12 + q_1^5q_2/120 + q_1^7/5040, \quad \dots \quad (\text{A.1})
 \end{aligned}$$

and

$$\begin{aligned}
 A_8 &= q_8 + q_7q_1 + q_6q_2 + q_5q_3 + q_4^2/2 + q_6q_1^2/2 + q_5q_2q_1 \\
 &\quad + q_4q_3q_1 + q_4q_2^2/2 + q_2q_3^2/2 + q_1^3q_5/6 + q_1^2q_3^2/4 \\
 &\quad + q_4q_2q_1^2/2 + q_1q_2^2q_3/2 + q_2^4/24 + q_1^4q_4/24 + q_1^3q_3q_2/6 \\
 &\quad + q_1^2q_3^2/12 + q_1^5q_3/120 + q_1^4q_2^2/48 + q_1^6q_2/720 + q_1^8/40320
 \end{aligned}$$

where

$$q_r = \frac{(-1)^{r-1}}{r(r+1)} \left[\sum_{i=1}^p B_{r+1} \left(\frac{\mu-i}{2} \right) - \sum_{i=1}^q \{B_{r+1}((\mu-1)p_i/2)\} / p_i^r \right],$$

$B_r(h)$ being the Bernoulli polynomial of degree r and order one.

(B) The coefficients R_i in (3.5) satisfy the following recurrence relation

$$\sum_{j=0}^i R_{i-j} C_{i-jj} = A_i, \quad R_0 = 1 \text{ and } C_{i0} = 1$$

where A_i are given in (A.1) above. C_{ij} are determined in terms of d_{ij} by replacing A_j by d_{ij} and q_j by C_{ij} in (A.1) above where

$$d_{ij} = \frac{(-1)^{j-1}}{j(j+1)} [B_{j+1}(0) - B_{j+1}(v+i)],$$

$B_j(h)$ being the Bernoulli polynomial and $v = (p^2 + p - 2q)/4$.

(C) The coefficients B_j occurring in (3.10) are obtained in terms of D_j by replacing A_j by B_j and q_j by D_j in (A.1) above where

$$D_j = \frac{(-1)^{j-1}}{j(j+1)} \left[\sum_{i=1}^p B_{j+1} \left(\frac{U-i}{2} \right) - \sum_{i=1}^q B_{j+1} ((U-1)p_i/2)/p_i^j \right].$$

Acknowledgment. The author is very grateful to the referee for suggesting corrections and improvements to an earlier version of this paper.

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Paper received : November, 1978.

Revised : July, 1979.